



Western Australian Certificate of Education Examination, 2010

Question/Answer Booklet

OUTDOOR EDUCATION

Stage 3

Please place your student identification label in this box

Student Number: In figures

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In words

Time allowed for this paper

Reading time before commencing work: ten minutes

Working time for paper: three hours

Materials required/recommended for this paper

To be provided by the supervisor

This Question/Answer Booklet

Multiple-choice Answer Sheet

To be provided by the candidate

Standard items: pens, pencils, eraser, correction fluid/tape, ruler, highlighters

Special items: navigational baseplate compass

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of exam
Section One: Multiple-choice	20	20	30	20	10
Section Two: Short answer	8	8	60	79	40
Section Three: Extended answer	3	2	90	40	50
Total					100

Instructions to candidates

1. The rules for the conduct of Western Australian external examinations are detailed in the *Year 12 Information Handbook 2010*. Sitting this examination implies that you agree to abide by these rules.
2. Answer the questions according to the following instructions.

Section One: Answer all questions on the separate Multiple-choice Answer Sheet provided. For each question shade the box to indicate your answer. Use only a blue or black pen to shade the boxes. If you make a mistake, place a cross through that square, do not erase or use correction fluid, and shade your new answer. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

Sections Two and Three: Write answers in this Question/Answer Booklet.

3. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.
4. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
 - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
 - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question(s) that you are continuing to answer at the top of the page.

Section One: Multiple-choice**10% (20 Marks)**

This section has **20** questions. Answer **all** questions on the separate Multiple-choice Answer Sheet provided. For each question shade the box to indicate your answer. Use only a blue or black pen to shade the boxes. If you make a mistake, place a cross through that square, do not erase or use correction fluid, and shade your new answer. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

Suggested working time: 30 minutes.

1. Maslow's 'Needs Hierarchy Theory' defines various levels of motivation. An individual at the level of 'Safety Needs' would be best described as seeking
 - (a) basic necessities of life.
 - (b) affiliation with a group.
 - (c) security through clear rules and expectations.
 - (d) fulfilment of personal potential.

2. The SMART Principle is often used in expedition planning when setting program, individual and group goals. SMART goals are best defined as
 - (a) specific, measurable, achievable, realistic and timely.
 - (b) specific, motivational, acceptable, relevant and trackable.
 - (c) significant, meaningful, achievable, realistic and timely.
 - (d) specific, measurable, acceptable, relevant and timely.

3. With respect to logistical planning, the construction of an expedition manual should include
 - (a) emergency procedures, participant information, support crew and transport.
 - (b) participant information, transport, accommodation and menu planning.
 - (c) accommodation, emergency procedures, transport and communication.
 - (d) transport, accommodation, participant information and support crew.

4. The Western Australian Adventure Activity Standards (AAS) were designed for the conduct of group outdoor adventure activities. The AAS provide a
 - (a) range of training opportunities.
 - (b) benchmark of minimum industry requirements.
 - (c) series of tools to promote reflective practice.
 - (d) rigid set of standard operating procedures.

5. Which strategy would be the **most** effective to assist others in the development of a new outdoor specific practical skill?
 - (a) give detailed description and demonstration
 - (b) offer feedback and encouragement
 - (c) provide a selection of appropriate resources
 - (d) facilitate open ended questioning

See next page

6. Analysis of an outdoor program with an educational focus would result in what outcome for the individual?
- (a) improvement in self esteem
 - (b) increased use of leisure time
 - (c) improvement in physical appearance
 - (d) increased understanding of the environment
7. When using roping skills related to an outdoor activity, the **most** important consideration for a knot is that it should be
- (a) recognisable.
 - (b) easily able to be undone.
 - (c) appropriate for the specific purpose.
 - (d) constructed quickly.
8. What is the most important interpersonal skill to use during a debrief?
- (a) Create a supportive atmosphere by being one of the group.
 - (b) Ask open ended questions to guide recognition of the learning experience.
 - (c) Encourage all group members to contribute in an open forum.
 - (d) Establish rules such as 'no put-downs' and 'listen to each other'.
9. Which strategy is **best** used by a facilitator to assist the development of personal skills in others?
- (a) creative writing
 - (b) solo time
 - (c) self-appraisal
 - (d) guided reflection
10. When creating a questionnaire to assist in the assessment of interpersonal skills in others, it is important that the questions
- (a) relate clearly to the individual or group goals.
 - (b) relate to the specific outdoor activity.
 - (c) require answers longer than one or two words.
 - (d) highlight individual opinions about the program.
11. Why is it important to use 'I' messages, rather than accusatory 'you' messages when dealing with conflict?
- (a) They make certain group members always listen to the other person.
 - (b) They calm down group members to feel comfortable to make themselves heard.
 - (c) They ensure that group members speak up without others putting them down.
 - (d) They enable group members to express their thoughts in an assertive, non-threatening way.

12. Which instructional method is used to assist in the facilitation of group performance?
- (a) journal writing
 - (b) observation
 - (c) team building
 - (d) self appraisal
13. A facilitator of a debrief session could assess its effectiveness by
- (a) observing how the group members adhered to the rules set for group operation.
 - (b) asking how well the group members achieved their personal goals.
 - (c) acknowledging how well the group members transferred learning into everyday life.
 - (d) using a reflection activity such as role playing or drawing.
14. Greenpeace and the Wilderness Society are two organisations that contribute to changing human-nature relationships. What is the **major** difference between these organisations?
- (a) method of action
 - (b) political lobbying
 - (c) environmental campaigns
 - (d) country of origin
15. Which example **best** describes an environmental management strategy used in the past by indigenous Australians?
- (a) hunting and gathering
 - (b) setting fish traps
 - (c) hunting with boomerangs
 - (d) lighting fires to enhance resources
16. During a bushwalk on the Bibbulmun Track, you think a cold front might be approaching. Which one of the following set of observations would confirm your opinion?
- (a) The air temperature is decreasing, the wind is from the northwest and cirrus clouds are visible.
 - (b) The air temperature is steady, the wind is from the south, cumulonimbus clouds fill the sky and some thunderstorms are evident to the north.
 - (c) The air temperature is decreasing rapidly, the wind is calm and shifting, there are no clouds visible but it looks hazy.
 - (d) The air temperature is decreasing, the wind is from the south and fog is rolling in.
17. Climate change has contributed to an extended drought in many parts of Australia. Which of the following responses **best** describes an example of an individual water conservation management strategy?
- (a) protecting catchment areas by prohibiting recreational use
 - (b) developing of desalination plants to produce fresh water from sea water
 - (c) diverting water from one catchment area to another
 - (d) installing domestic appliances and fixtures that use less water

18. Which one of the following practices is an example of the Leave No Trace principle 'travel and camp on durable surfaces'?
- (a) leave rocks, plants and other natural objects as you find them
 - (b) if walking off tracks, spread the group out
 - (c) set up your stove on a flat site
 - (d) schedule your trip to avoid times of high use
19. In which direction do winds travel around an anticyclone in the southern hemisphere?
- (a) clockwise and slightly in toward the centre of the system
 - (b) clockwise and slightly out from the centre of the system
 - (c) anti-clockwise and slightly in toward the centre of the system
 - (d) anti-clockwise and slightly out from the centre of the system
20. Leave No Trace is an environmental ethic for outdoor recreation enthusiasts. It is best understood as
- (a) a set of rules and procedures that must be used during outdoor recreation.
 - (b) a set of principles that you apply to minimise your impact and preserve the environment.
 - (c) guidelines that permit low impact visits but not camping at a site.
 - (d) regulations imposed by the Department of Conservation and National Park Authorities.

End of Section One

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See next page

Section Two: Short answer

40% (79 Marks)

This section has **eight (8)** questions. Answer **all** questions. Write your answers in the spaces provided.

Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.

- Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
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Suggested working time: 60 minutes.

Question 21

(16 marks)

Your group has been hiking along the coastline in the Mount Manypeaks region. Unsure of your exact location, you decide to stop to perform a triangulation.

- (a) You can visually identify three distinct hills. Mark each hill with a small cross (x) and label each of the **three (3)** hills on the map on page 10. (3 marks)

- (i) Hill 1 – GR 297375
- (ii) Hill 2 – GR 344344
- (iii) Hill 3 – GR 280358

- (b) Use the North Points diagram on page 11 to calculate the magnetic variation. Show your working clearly and round your answer to the nearest whole degree. (3 marks)

- (c) Using your compass, you take bearings to each of these hills. These bearings are recorded below. Convert the **three (3)** bearings to back bearings, including the allowance for magnetic variation. Show your working clearly. (6 marks)

- Hill 1 – Bearing 57°
- Hill 2 – Bearing 112°
- Hill 3 – Bearing 196°

Hill 1

Hill 2

Hill 3

- (d) Use these back bearings to accurately draw the lines of triangulation on the map on page 10 to complete the task. (3 marks)
- (e) Identify your location accurately with a **six (6)** digit grid reference. (1 mark)

Mount Manypeaks map



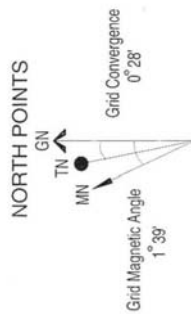


TOPOGRAPHIC MAP
1:100 000 SCALE (1 cm to 1 km)

2528
EDITION 1

Manypeaks Special

WESTERN AUSTRALIA



True North, Grid North and Magnetic North are shown diagrammatically for the centre of this map. Magnetic North is correct for 2002 and moves westerly by 6° 07" annually. A Grid Bearing is obtained by subtracting the Grid Magnetic Angle from the Magnetic Bearing.

SCALE 1: 100 000

10'



118°13'30"

Legend

Built-up area; Dual carriageway road		Lake, perennial; Watercourse, perennial		Homestead; Building; Ruin; Yard	
Principal road; Route marker: National; State ..		Lake, intermittent; Watercourse, intermittent ...		Fence; Levee; Quarry	
Secondary road; Embankment; Cutting		Lake, mainly dry; Watercourse, mainly dry		Mine; Windpump	
Minor road; Road bridge		Swamp or marine swamp		Contour with value; Depression contour	
Vehicle track (unsealed)		Land subject to inundation		Horizontal control point; Spot elevation	
Gate; Stock grid		Bore or well; Spring; Watertank or small dam ..		Powerline; Heliport	
Multiple track railway; Station or siding		Wreck, bare or awash; submerged		Sand; Sand dunes	
Single track railway; Bridge; Tunnel		Foreshore flat; Lighthouse		Sand ridges; Cliff; Razorback ridgeline	
Light railway or tramway		Reef; Shoal; Mangrove		Forest/scrub, scattered; medium; dense	
State or Territory boundary		Rock, bare or awash; submerged; Breakwater ..		Pine plantation; Rainforest	
Prohibited area boundary		Jetty, wharf or pier; Saline coastal flat		Orchard (vineyard); Windbreak	
				Reserve area boundary	

Question 22**(12 marks)**

Effective equipment management before, during and after an expedition is a vital consideration for all outdoor experiences. Describe **three (3)** strategies for effective equipment management in **each** of the following:

(a) Storage (3 marks)

(b) Maintenance (3 marks)

(c) Care (3 marks)

(d) Records

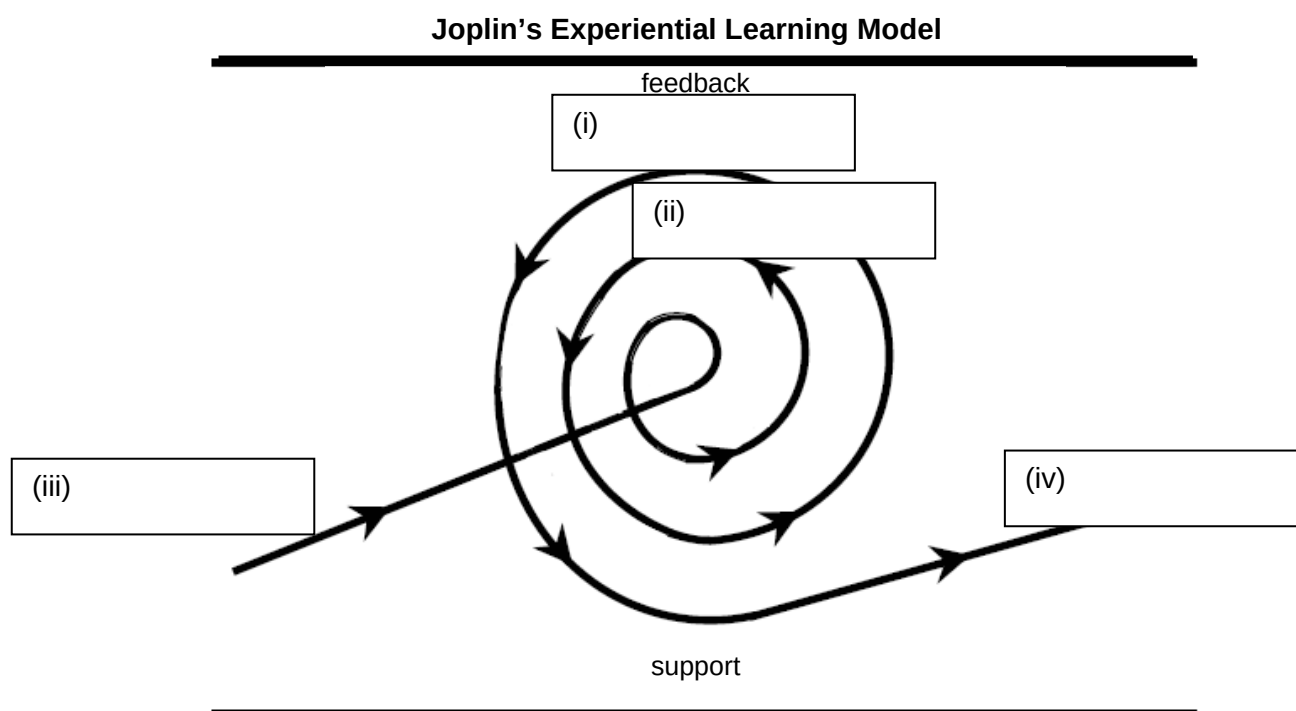
(3 marks)

Question 23

(14 marks)

(a) Label the diagram of Joplin's Experiential Learning Model.

(4 marks)

(b) Describe any **two (2)** stages of Joplin's Experiential Learning Model.

(4 marks)

- (c) Name and explain **three (3)** skills the facilitator of an outdoor activity would demonstrate to assist the experiential learning process. (6 marks)

Question 24**(8 marks)**

Identify **four (4)** verbal and **four (4)** non-verbal communication skills that you would assess in an Interpersonal Skills Peer Review/Rating Appraisal.

Verbal Communication

One:

Two:

Three:

Four:

Non-verbal Communication

One:

Two:

Three:

Four:

Question 25

(5 marks)

Complete the Leadership Theories table.

Leadership theory	Characteristics/description
Behavioural/charismatic and heroic	(i) _____ _____ _____ _____ _____
(ii) _____	Leaders are born, not made. This approach differentiates leaders from non-leaders by physical and psychological attributes.
(iii) _____	Models of leadership that take into account the leader, the followers and the situation. Different skills are needed for different situations, so the person with the correct skills becomes the leader.
Contingency	(iv) _____ _____ _____ _____ _____
Dynamic	(v) _____ _____ _____ _____ _____

Question 26**(8 marks)**

The term 'fire-stick farming' has been used to describe the traditional practice by indigenous people of periodically burning country.

- (a) What was the purpose of fire-stick farming? (2 marks)

- (b) How has fire-stick farming helped to shape the Australian environment we experience today? (3 marks)

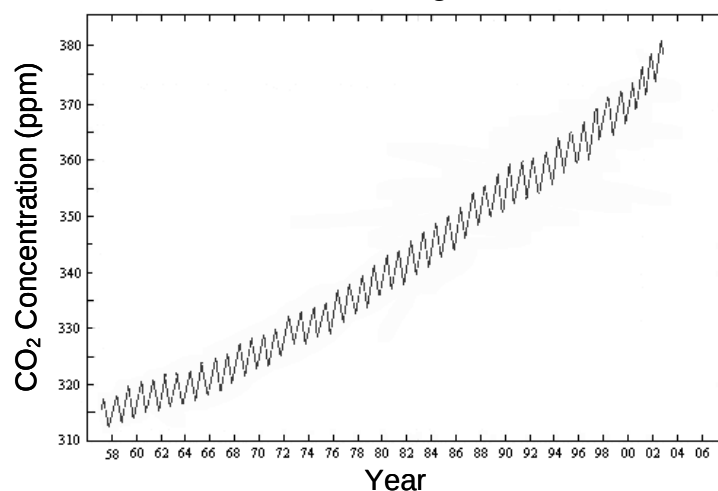
- (c) What modern environmental practices maintain the role of the indigenous people's fire-stick farming? (3 marks)

Question 27

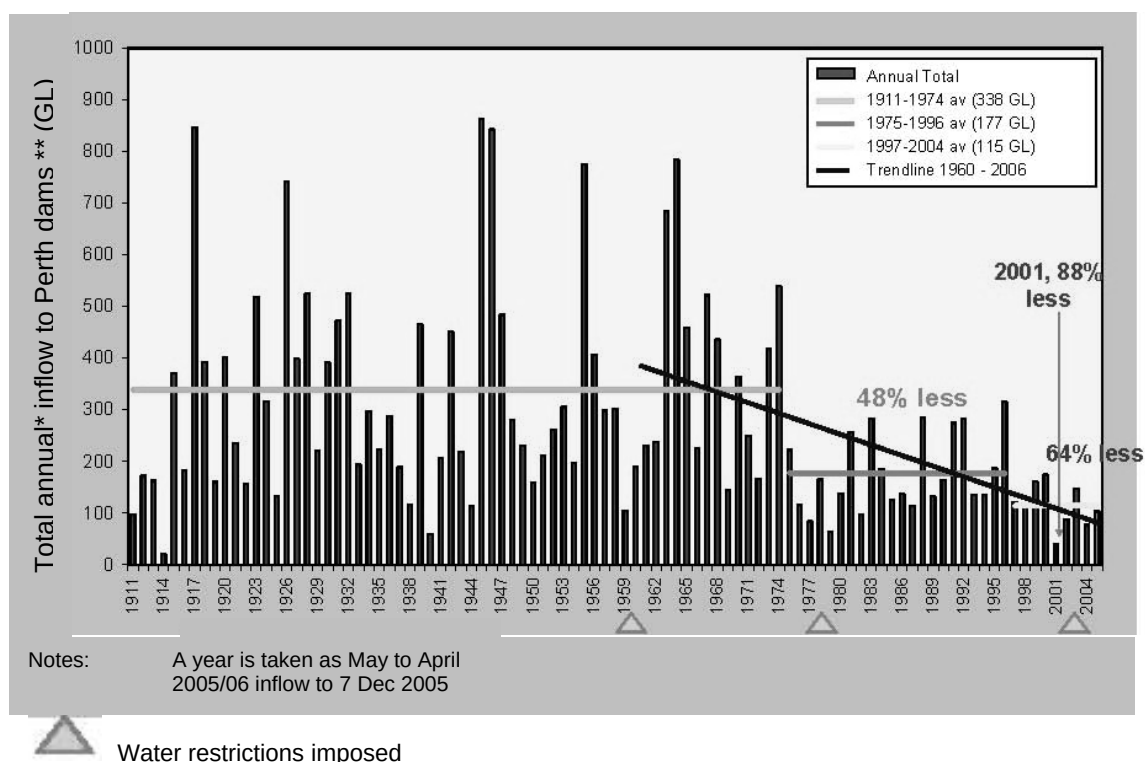
(8 marks)

The following two graphs provide data on carbon dioxide (CO₂) levels in the atmosphere and water inflows to Perth's catchment areas.

The Keeling Curve



Impact on water availability – WA reduced inflows to dams



- (a) What are the **two (2)** main human-induced causes of changes to CO₂ levels? (2 marks)

- (b) What are **three (3)** consequences of increased CO₂ levels on the weather experienced in Western Australia? (3 marks)

- (c) With reference to the data presented and your own experiences, suggest how people in Western Australia may change their environmental behaviors in the coming decade to help reduce CO₂ emissions. (3 marks)

Question 28

(8 marks)

- (a) Describe **four (4)** types of weather observations that would be useful in predicting future weather patterns while on an expedition. (4 marks)

- (b) Below is a set of weather observations taken on a bushwalking expedition. Complete the table detailing the actions required for the safe conduct of the expedition. (4 marks)

Weather observation taken at 3:30 pm	Actions required
Wind is calm with some strong erratic gusts from the north.	
Building cumulous with spreading anvil tops over the hills.	
Temperature seems to be dropping rapidly.	
No rain but it is hazy to the north.	

End of Section Two

See next page

Section Three: Extended answer**50% (40 Marks)**

This section contains **three (3)** questions. You must answer **two** questions. Write your answers in the spaces provided.

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Suggested working time: 90 minutes.

Question 29 – Outdoor experiences**(20 marks)**

Use an expedition that you have conducted to highlight your understanding of the adventure experience paradigm. Name and describe each stage of the paradigm. Define the terms 'risk' and 'competence' and explain the relationship that exists between them. Analyse the physical and emotional state of different participants with respect to the achievement of a peak adventure experience.

Question 30 – Self and others**(20 marks)**

Facilitator



Original image altered by Curriculum Council for copyright reasons.

Group facilitation

- (a) Use your experience gained from participating in an expedition to identify the interpersonal skills and facilitation techniques being used by the facilitator in the photo above. (5 marks)
- (b) Name and define each of the **three (3)** debriefing phases. Use your personal experience to outline the facilitation methods used in each phase. Highlight how the design and implementation of each phase assisted group members in their learning on the expedition. (15 marks)

See next page

Question 31 – Environmental awareness

(20 marks)

Reflect on an environmental interpretation activity that you have facilitated and considered to be effective. Describe it clearly and explain its goals and suitability for the participants. Analyse and evaluate the effectiveness of the activity in achieving its goals, including aspects relating to how and where it was conducted.

Your answer must show clear links between the activity chosen, its goal and its suitability for the group and also demonstrate your ability to reflect on its effectiveness as a whole and the effectiveness of the processes involved.

Question number: _____

[illegible]

End of questions

Question number: _____

[illegible]

Question number: _____

[illegible]

Question number: _____

[illegible]

Question number: _____

[illegible]

Question number: _____

[illegible]

Question number: _____

[illegible]

Question number: _____

[illegible]

Spare answer page:

Question number: _____

[illegible]

Spare answer page:

Question number: _____

[illegible]

ACKNOWLEDGEMENTS

Section Two: Short answer

- Question 21** NATMAP. (n.d.). *Manypeaks special Western Australia* [Topographic map: 2528 Ed. 1]. Geoscience Australia.
Permission to reproduce given by the Commonwealth of Australia (Geoscience Australia).
- Question 23** Adapted from: Joplin, The five-stage experiential learning model, 1981. In T. Grady Roberts. (2006). *A Philosophical Examination of Experiential Learning Theory for Agricultureal Educators* [Journal of Agricultural Education 20 Volume 47 (1)]. Texas A & M University, p. 20.
- Question 27** Graph adapted from: Keeling, C. (Dr.). The Keeling Curve. In Hayes-Bonanan, J. (2008). *An Inconvenient Geography: Global Climate Change*. Retrieved August, 2010, from webhost.bridgew.edu/jhayesboh/climate/index.htm.
- Graph adapted from: Gill, J. (Dr.). (2006). *Securing our water future in a Drying climate* (p. 6). Water Corporation. Retrieved August, 2010, from www.watercorporation.com.au/_files/Presentations/20060510.pdf

Section Three: Extended answer

- Question 30** Photograph adapted from: St Brigid's College. (n.d). *Group facilitation*.
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